The Global Web of Bonds: How Bonds Hand Over Across Crystal Networks



Theorem: for ten classical two-kind crystal site sets, and one three-kind set, the weighted Voronoi (power) diagram is computed exactly along the whole weight line on which both kinds have cells: every flip, the bonds on every interval, and every point where only unlike bonds exist. The two hand-overs of the DICTONIC loops, gradual and with a single switch, both recur.

DICTO (artist, Osaka, Japan) · contact@dictospheres.com · https://dictospheres.com
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The ideas are DICTO's: the DICTONIC SOLIDS and the DICTONIC BONDS and their names, the web of bonds, and DICTO's idea, proposed on 25 October 2026, that "the dictonic cycle branches could show up elsewhere in geometric networks in slightly different guises"; computation, exact proofs and drafting with Claude (Anthropic).

In short. Paper #52 [3] showed that DICTO's two loops of space-filling pairs are one fixed crowd of points of two kinds, with one number, a weight, slid between the kinds; the bonds (one line for each face two pieces share) form a web. In the golden loop the two like bonds hand over gradually; in the Delian loop they hand over all at once, at the octet, where only bonds between unlike pieces are left. DICTO proposed that these branches turn up elsewhere in slightly different guises. We took eight more crystal arrangements of two kinds of points (three in the plane, five in space) and one of three kinds, and slid the weight through every value at which both kinds still have a piece. For each we found, exactly, every point where the pattern changes, which bonds exist in between, and whether there is a moment with only unlike bonds. The proposal holds: the hand-over with a single switch recurs in the square lattice with its centres, in the honeycomb with its hexagon centres (the switch is the trihexagonal tiling) and in rock salt; the gradual hand-over recurs in zinc blende, in the holes of hexagonal close packing, and in the A15 and C15 alloy structures. In every two-kind network the bond between unlike pieces never breaks, and we prove that it never can. Much of this sits inside known theory, and we say where.

Main theorem. For each of the ten two-kind networks of Table 1, with weight difference $\lambda = w_A - w_B$: both kinds have cells exactly on the open range of Table 2; the face lattice changes exactly at the listed rational flips and is constant between them; every bond type lives on one interval, given exactly in Table 2, and the unlike bond lives on the whole open range; a parameter at which both kinds have cells and only unlike bonds exist occurs exactly at $\lambda = 0$ (square + centres), $-1/2$ (honeycomb + centres), 0 (rock salt) and $-1/8$ (fluorite, the octet), and nowhere in the other six. For the three-kind perovskite the coexistence region is the open rectangle $-2 < \lambda < 2$, $-1 < \mu < 1$, cut by three straight walls into five chambers, and no point of it has only unlike bonds. All results are exact (648 exact and float checks, none failed). The named tilings at the stops follow from a classical rule (Aurenhammer; Ash and Bolker); the A15 walk up to its first flip is Weaire

and Phelan's (with Kashyap and Neuhoff, as reported by Sloane and Vaishampayan); the switch points are walls in the sense of Gelfand, Kapranov and Zelevinsky.

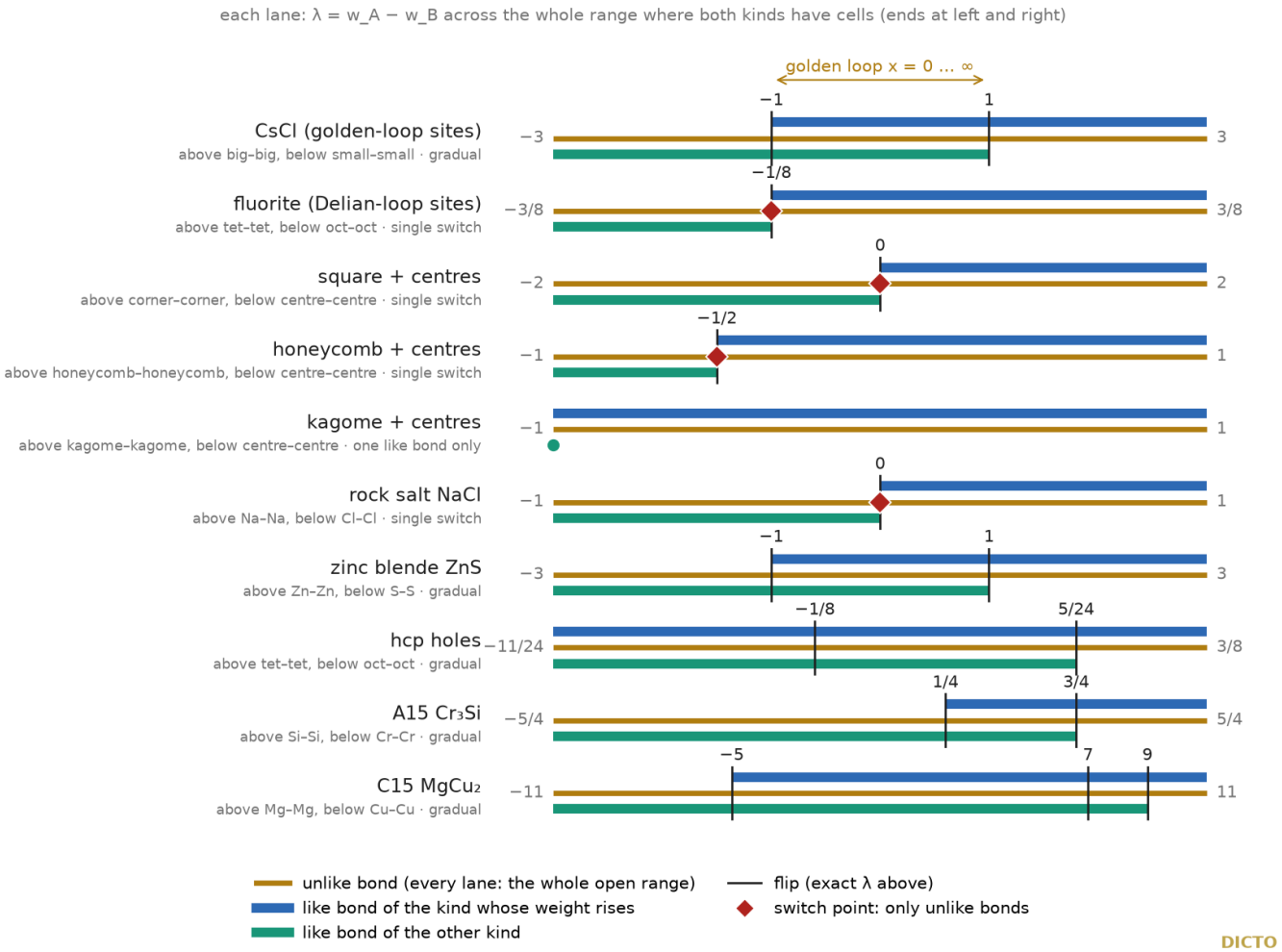


Figure 1. The ten walks of the Main theorem, each lane spread over its own coexistence range (ends at left and right, in the network's units, Table 1). Gold: the unlike bond, alive on the whole open range in every lane. Blue: the like bond of the kind whose weight rises with λ; green: the other like bond. Black ticks: the flips, with their exact λ. Red diamonds: the switch points, where only unlike bonds exist. Where blue and green overlap, the hand-over is gradual; where they meet at a diamond, it has a single switch. The golden loop of paper #48 [1] is exactly the overlap (−1, 1) of the CsCl lane, with its ends; the Delian loop of paper #51 [2] is the whole fluorite lane. Kagome + centres has only one like bond (the centre–centre bond appears only at the end λ = −1, as a dot).

1. Where it started

> *Two loops of DICTO's showed two ways in which bonds can hand over. DICTO proposed that the same branches appear in other networks. We walked the weight through ten networks to find out.*

Paper #48 [1] found the golden pair (the Icosaxes and the Unborn Axes) in a one-parameter family of space-filling truncated octahedra; paper #51 [2] found the Delian bond (Samson and Delilah) in a one-parameter family between Kepler's rhombic dodecahedron and the cube. DICTO calls the solids the DICTONIC SOLIDS and the pairs the DICTONIC BONDS. Paper #52 [3] proved that both loops are power diagrams of fixed sites of two kinds, the caesium chloride sites and the fluorite sites, with one weight slid between the kinds, and that the bonds in the golden loop hand over gradually while those in the Delian loop hand over at the octet. On 25 October 2026 DICTO proposed that these branches could show up elsewhere in

geometric networks in slightly different guises. This paper reports the exact study that tests that proposal [S].

2. The set-up

> *Fix the points; give each kind a weight; let the difference of the weights run. Each piece is the region nearer, in the weighted sense, to its own point than to any other.*

Power diagram. Given sites c_i with weights w_i, the power cell of c_i is the set of points p with $|p - c_i|^2 - w_i \le |p - c_j|^2 - w_j$ for all j [6, 7]. The cells are convex and fill space; two of them meet, if at all, in the power bisector of their sites. Two sites are *bonded* when their cells share a face of full dimension (an edge in the plane, a polygon in space); the bonds are the edges of the dual weighted-Delaunay (regular) subdivision [6]. A bond is *unlike* when it joins sites of different kinds and *like* otherwise.

The networks. Each network is a periodic set of sites of two kinds A and B, with $w_A = \lambda$ and $w_B = 0$, so $\lambda = w_A - w_B$ (Table 1). The *coexistence range* is the set of λ at which both kinds have cells of positive volume. A *flip* is an interior value of λ at which the face lattice of the diagram changes. Following #52, the two like bonds hand over *gradually* when both are alive on a common open interval, and *with a single switch* when one dies exactly where the other is born. A *switch point* is a value of λ at which both kinds have cells and every bond is unlike: the octet is the switch point of the Delian loop.

Network	Kinds A, B	Sites and units
CsCl (golden-loop sites)	big, small	big $2\mathbb{Z}^3$, small $2\mathbb{Z}^3 + (1, 1, 1)$: half cube edge 1. The golden loop's $x \in [0, \infty]$ is $\lambda = (x - 1)/(x + 1)$
fluorite (Delian-loop sites)	tet, oct	fcc ("oct") and all its tetrahedral holes ("tet"); octet edge 1, the app's units. $\lambda = (3/8)(2t - 1)$
square + centres	corner, centre	corners $2\mathbb{Z}^2$, centres $2\mathbb{Z}^2 + (1, 1)$: square edge 2
honeycomb + centres	honeycomb, centre	the triangular lattice of edge 1, split 2 : 1 into the honeycomb points and the hexagon centres
kagome + centres	kagome, centre	the triangular lattice of edge 1, split 3 : 1 into the kagome points and the hexagon centres
rock salt NaCl	Na, Cl	two face-centred cubic lattices; Na–Cl distance 1
zinc blende ZnS	Zn, S	fcc (Zn) and one set of its tetrahedral holes (S); cubic cell edge 4, Zn–S distance √3, Zn–Zn distance 2√2
hcp holes	tet, oct	the octahedral and tetrahedral holes of ideal hexagonal close packing, the cell centres of the gyrated tetrahedral–octahedral honeycomb (used in DICTO's papers #37a and #40 [4, 5]); hcp edge 1
A15 Cr_3Si	Si, Cr	the A15 (β-tungsten) sites; cubic cell edge 2
C15 Laves $MgCu_2$	Mg, Cu	the C15 sites; cubic cell edge 8
perovskite ABX_3 (three kinds)	A, B, X	A cube corners, B body centre, X face centres; cubic cell edge 2 (A–X distance √2, B–X distance 1); $\lambda = w_A - w_X$, $\mu = w_B - w_X$

Table 1. The networks and their units [S, README "Units"]. Hexagonal sites are handled exactly in a 60° basis with a rational metric; a power diagram in such a metric is an affine image of the Euclidean one, with the same combinatorics and the same weights.

3. Main theorem

> *For every network the whole walk is known exactly: where both kinds live, where the pattern flips, which bonds exist in between, and where only unlike bonds are left.*

Main theorem (the census). Let N be one of the ten two-kind networks of Table 1, with $\lambda = w_A - w_B$.
(i) *Range.* Both kinds have cells of positive volume exactly for λ in the open interval of Table 2. At its lower end every A cell is a single point, at its upper end every B cell is.
(ii) *Flips.* The face lattice of the power diagram is constant on each open interval between consecutive values in the "flips" column of Table 2, and differs on the two sides of each of them. All flips are rational. Appendix C lists, for every interval and every stop, the face vector of every cell and its bonds by neighbour kind.
(iii) *Bond lives.* Each bond type is present exactly on the single interval of Table 2 (a bracket marks an end included). In particular, in all ten networks the unlike bond is present on the whole open range and only there.
(iv) *Switch points.* A switch point exists exactly in four networks: λ = 0 in square + centres, λ = −1/2 in honeycomb + centres, λ = 0 in rock salt and λ = −1/8 in fluorite. There the dual cells are squares (2 corner + 2 centre sites), rhombi (2 honeycomb + 2 centre sites), cubes (4 Na + 4 Cl) and rhombic dodecahedra (6 oct + 8 tet), and every piece has only unlike bonds: 4 and 4; 3 per honeycomb piece and 6 per centre piece; 6 and 6; 8 per oct piece and 4 per tet piece. In the other six networks no value of λ in the range, flip or not, has only unlike bonds.
(v) *Three kinds.* For the perovskite sites, all three kinds have cells of positive volume exactly on the open rectangle $-2 < \lambda < 2$, $-1 < \mu < 1$ (its closure has area 8). It is cut by the three straight walls λ = 0, λ + 2μ = 0 and λ + μ = 1 into five chambers of constant face lattice. Each wall is the birth or death line of exactly one bond type: A–A is present for λ > 0, A–B for λ + 2μ > 0, X–X for λ + μ < 1. A–X and B–X are present in every chamber, B–B in none, and no point of the open rectangle (chamber, wall or corner) has only unlike bonds.

Network	Range	Flips	A–A present	B–B present	Hand-over	Switch point
CsCl	(−3, 3)	−1, 1	big–big (−1, 3]	small–small [−3, 1)	gradual, both on (−1, 1)	none
fluorite	(−3/8, 3/8)	−1/8	tet–tet (−1/8, 3/8]	oct–oct [−3/8, −1/8)	single switch	−1/8 (octet)
square + centres	(−2, 2)	0	corner–corner (0, 2]	centre–centre [−2, 0)	single switch	0 (square tiling)
honeycomb + centres	(−1, 1)	−1/2	honeycomb–honeycomb (−1/2, 1]	centre–centre [−1, −1/2)	single switch	−1/2 (trihexagonal)
kagome + centres	(−1, 1)	none	kagome–kagome (−1, 1]	centre–centre only at −1	none: one like bond	none
rock salt	(−1, 1)	0	Na–Na (0, 1]	Cl–Cl [−1, 0)	single switch	0 (cubes)
zinc blende	(−3, 3)	−1, 1	Zn–Zn (−1, 3]	S–S [−3, 1)	gradual, both on (−1, 1)	none
hcp holes	(−11/24, 3/8)	−1/8, 5/24	tet–tet (−11/24, 3/8]	oct–oct [−11/24, 5/24)	gradual, both on (−11/24, 5/24)	none
A15	(−5/4, 5/4)	1/4, 3/4	Si–Si (1/4, 5/4]	Cr–Cr [−5/4, 3/4)	gradual, both on (1/4, 3/4)	none
C15	(−11, 11)	−5, 7, 9	Mg–Mg (−5, 11]	Cu–Cu [−11, 9)	gradual, both on (−5, 9)	none

Table 2. The ten walks [S, README "Summary table" and "Lives of the bonds"; relay-summary.json]. λ in the units of Table 1. The unlike bond is present exactly on the open range in every row. The ends of the like-bond lives are closed where the other kind's cells have shrunk to points but the like faces survive.

Why: along the weight line every corner of every cell moves on a straight line, so each corner stays a corner until one exact rational value; walking from one such value to the next covers the whole line.

Proof (computer-assisted, exact). Each site set has rational coordinates in a rational metric G. The power inequality between sites i and j is $n_{ij} \cdot p \le b_{ij}(\lambda)$ with n_{ij} rational and b_{ij} affine in λ with rational coefficients. A corner is the solution of d such equalities, so it is $p(\lambda) = A^{-1}b(\lambda)$, affine in λ with rational coefficients, and the slack of every other inequality at that corner is affine in λ. Hence, at a rational sample λ_0, the face lattice is constant on the open interval on which no slack that is positive at λ_0 reaches 0; its ends are rational [S, scripts/engine.py]. Floats only choose candidate corners; every corner is then solved and tested in exact fractions. The walk starts at $\lambda = 0$, finds the interval, analyses its end (the stop), steps just past the stop and repeats, checking that each new interval begins exactly at the previous stop [S, scripts/census.py]. On every interval: the cell volumes, as polynomials in λ, add up to the volume of the lattice cell identically (so no corner and no cell is missing); every face is found in both of its cells with the same measure; Euler's formula holds for every three-dimensional cell; and a distance bound shows that the finite list of sites cannot be cut by a farther site anywhere on the closed interval. At every stop the limit of the interval's corners equals the power cell computed from scratch at that exact λ. (i) Increasing λ only relaxes the inequalities of A cells against B sites, so every A cell grows and every B cell shrinks with λ; each kind therefore has cells on an interval, and the walk ends where a cell volume polynomial vanishes and the cell is checked to be a point. (ii) The interval ends are the flips; that the face lattice differs on the two sides of each flip is read off the interval records. (iii) The life of each bond type is read off the records of all intervals and stops, with a check that it is present on every record between its ends [S, scripts/relay.py]. (iv) All records are searched for parameters with both kinds alive and only unlike bonds; the dual cells and bond counts are read off the corner incidences. (v) For the perovskite, every slack is affine in (λ, μ); exact chambers are found and the plane is explored by stepping across every wall, and every interior wall segment is checked to be covered by chambers on its other side [S, scripts/perovskite.py]; on each of the five open interior wall segments no other wall crosses, so the face lattice is constant along it, and one exact point (its midpoint) shows a like bond there [S, scripts/walls.py]. An independent float computation at three random parameters inside every interval gives the same bond counts [S, scripts/crosscheck.py]. Both DICTONIC loops of #52 are reproduced exactly, including the Delian flip at $t = 1/3$ and its two equal-edged stops. In all: 543 checks on the walks, 41 on the bond lives (including 9 regressions against #52's study), 47 on the perovskite, 12 float and dual-cell cross-checks, 5 on the perovskite walls; 648 PASS, 0 FAIL (Appendix B). ∎

4. Corollaries

> *The theorem places DICTO's two loops inside larger families, shows that every switch point is a meeting of two ways to cut the same dual cell, and finds classical tilings at the named stops.*

Corollary 1 (the DICTONIC loops inside the census). In half-cube-edge units the golden loop of #48 is $\lambda = (x - 1)/(x + 1)$, so $x \in [0, \infty]$ covers exactly $[-1, 1]$: the gradual-hand-over interval of CsCl, from flip to flip. Its ends $x = 0$ and $x = \infty$ are those flips, where the cells are octahedra and cuboctahedra; beyond them, on $(-3, -1)$ and $(1, 3)$, the CsCl family continues with truncated cubes and octahedra, ending in cubes at ± 3 [S, README]. The Delian loop of #51, $\lambda = (3/8)(2t - 1)$, covers the whole fluorite range, and its switch is the octet.

Corollary 2 (switch points are walls). In any two-kind network, every switch point is a wall: there every bond is unlike, so the edges of every dual cell join sites of different kinds,

while a simplex with three or more corners of two kinds always has an edge joining two of the same kind; hence no dual cell is a simplex. At each of the four switch points the dual cells are not simplices (Main theorem (iv)), and on the two sides they are cut in the two possible ways: the dual squares and rhombi into two triangles across one diagonal or the other (2 centre + 1 corner or 1 centre + 2 corner sites; the same for the honeycomb); the dual cubes of rock salt into four corner tetrahedra and one central tetrahedron (3 Cl + 1 Na and 4 Cl below 0, 1 Cl + 3 Na and 4 Na above); the rhombic dodecahedra of fluorite as in #52 [S, data/raw/*.json "dual"]. In the language of regular subdivisions [8, 9] each switch point is a wall where two triangulations of the same non-simplicial cell meet, and the switch is a flip in that sense. For square + centres and rock salt the wall is forced at $\lambda = 0$: with equal weights the sites form a lattice (a square lattice, the simple cubic lattice) whose Delaunay cells are squares and cubes; Voronoi's theory of lattice Delaunay subdivisions covers this case [10, 11]. The swap of the two diagonals in a quadrilateral is the elementary two-dimensional flip; the two cuts of the cube used here are its two "corner-cut" triangulations into five tetrahedra [9].

> **Proposition 1 (the unlike bond can never break).** In any face-to-face tiling of space by convex cells of two kinds A and B (periodic or not, locally finite), if both kinds have cells of positive volume then some A cell and some B cell share a facet. Hence, in every two-kind network, the unlike bond is present at every parameter at which both kinds have cells. With three or more kinds, every kind shares a facet with some cell of another kind: each kind keeps some unlike bond, but not necessarily with every other kind.

> *Why: the A cells and the B cells together fill space, so somewhere an A region must border a B region along a flat piece of wall.*

Proof. Let U_A be the union of the A cells; it is closed, as a locally finite union of closed sets. It has nonempty interior, and an interior point of a B cell lies in no A cell (cells have disjoint interiors), so U_A is a proper subset of space. Hence its boundary ∂U_A is nonempty, and space minus ∂U_A is the disjoint union of the interiors of U_A and of U_B, both nonempty, so it is disconnected. ∂U_A is a union of pieces of facets of A cells; if every piece had dimension at most $d - 2$ it could not disconnect space, so ∂U_A contains a relatively open piece of positive $(d - 1)$-measure of a facet of some A cell. The points just across that piece lie in B cells, finitely many near it, so one B cell meets that facet in a set of positive $(d - 1)$-measure; since the tiling is face-to-face, that set is a common facet, an A–B bond. With more kinds, apply the argument to one kind against the union of all others. ∎ This explains the perovskite: A–B dies on $\lambda + 2\mu = 0$, but A and B each keep their bonds with X. We found no statement of this elementary fact in the sources searched.

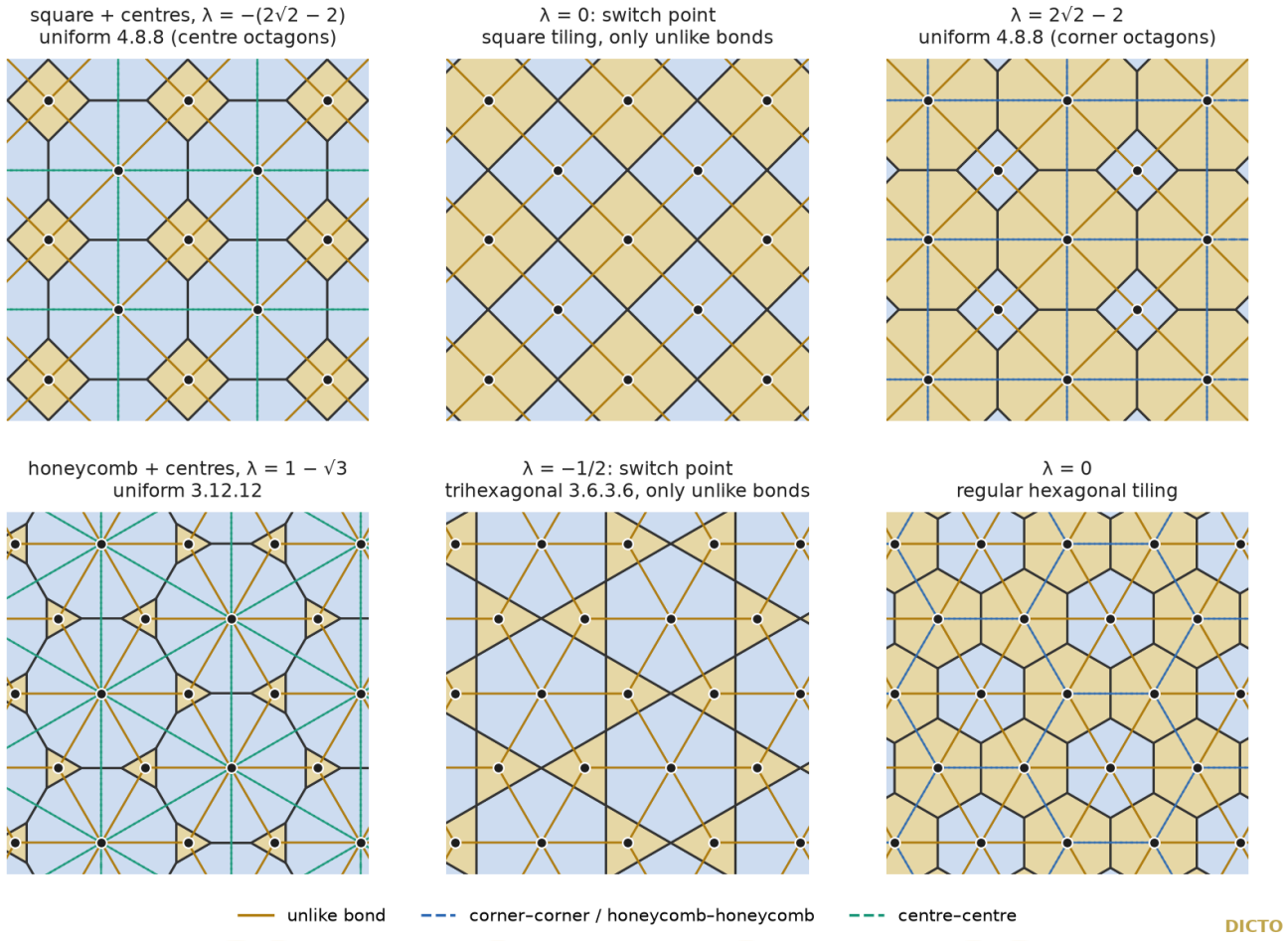


Figure 2. The two planar networks with a single switch, drawn at three stops each (float pictures of the exact walks; cells of the A kind gold, of the B kind blue; black dots: the sites of both kinds; bonds drawn between bonded sites). Top: square + centres at $\lambda = -(2\sqrt{2} - 2)$, 0 and $2\sqrt{2} - 2$. Bottom: honeycomb + centres at $\lambda = 1 - \sqrt{3}$, $-1/2$ and 0. In the middle column only unlike (gold) bonds are left; on its left the centre–centre bonds (green) are present, on its right the corner–corner or honeycomb–honeycomb bonds (blue). Each dual square (top) or rhombus (bottom) is cut by one like diagonal on each side of the switch.

Proposition 2 (classical: the circumsphere rule). If every tile of a face-to-face convex tiling has a circumscribed circle or sphere, then two tiles that share a facet have their facet in the radical line or plane of their two circles or spheres (its corners have power 0 with respect to both), and the tiling is the power diagram of the tiles' circumcentres with weights equal to the squared circumradii [6, 12, 13, 14]. Every uniform tiling and honeycomb has such tiles.

Corollary 3 (named tilings at the stops). Where a uniform tiling is a stop or a special point of a walk, its λ is the difference of the squared circumradii of the A tile and the B tile, and the census finds exactly that value [S, prior-art.md, find 2]: the uniform truncated square tiling 4.8.8 at $\lambda = \pm(2\sqrt{2} - 2)$ in square + centres (all edges equal; the octagons are the centre cells for $\lambda < 0$ and the corner cells for $\lambda > 0$); the uniform truncated hexagonal tiling 3.12.12 at $\lambda = 1 - \sqrt{3}$ and the trihexagonal tiling 3.6.3.6 at the switch point $\lambda = -1/2$ (one edge length, edge$^2 = 3/4$) in honeycomb + centres; the uniform truncated cubic honeycomb (truncated cubes and octahedra) at $\lambda = \pm(2\sqrt{2} - 1)$ in CsCl; the octet at $\lambda = -1/8$ in fluorite and the gyrated tetrahedral–octahedral honeycomb at $\lambda = -1/8$ in the hcp holes. These are instances of Proposition 2, not new tilings. Away from uniform tilings the census also finds exact stops the rule does not cover: in rock salt every edge of the tiling has one length at $\lambda = \pm(4 - 2\sqrt{3})$, where the larger cell has 6 squares and 12 hexagons (the face vector of the chamfered cube) and the smaller is a cube (Figure 3).

Corollary 4 (the octet in hexagonal stacking). In the hcp holes, whose diagram at $\lambda = -1/8$ is the gyrated tetrahedral–octahedral honeycomb (regular octahedra and tetrahedra; the gyrated octet of DICTO's #37a and one of the stackings of #40 [4, 5]), the octet's flip stays at $\lambda = -1/8$: there oct–oct bonds at squared distance 1 die and tet–tet bonds are born (at squared distance 1/2, as in fluorite, and also at 1). But $-1/8$ is not a switch point: across the gyrated layers each oct piece keeps 2 oct–oct bonds and each tet piece 1 tet–tet bond. Of these cross-layer bonds, tet–tet at squared distance 1/6 lives on the whole range $(-11/24, 3/8]$ and oct–oct at squared distance 2/3 dies only at a second flip, $\lambda = 5/24$ (Table 3). The dual cell at an atom at $\lambda = -1/8$ is the trapezo-rhombic dodecahedron (6 rhombi and 6 trapezoids), where fluorite has the rhombic dodecahedron (12 rhombi) [S, crosscheck-checks.txt].

Network	Bond family (type @ squared site distance)	Present
hcp holes	oct–tet @ 3/8	$(-11/24, 3/8)$
	oct–oct @ 1 (in-layer)	$[-11/24, -1/8)$
	tet–tet @ 1/2 and @ 1 (in-layer)	$(-1/8, 3/8]$
	oct–oct @ 2/3 (cross-layer)	$[-11/24, 5/24)$
	tet–tet @ 1/6 (cross-layer)	$(-11/24, 3/8]$
A15	Cr–Si @ 5/4	$(-5/4, 5/4)$
	Cr–Cr @ 1 and @ 3/2	$[-5/4, 3/4)$
	Si–Si @ 3	$(1/4, 5/4]$
	Si–Si @ 4	$(3/4, 5/4]$
C15	Mg–Cu @ 11	$(-11, 11)$
	Cu–Cu @ 8	$[-11, 9)$
	Mg–Mg @ 12	$(-5, 11]$
	Mg–Mg @ 32	$(7, 11]$

Table 3. Bond families where a bond type splits by distance [S, README "Lives of the bonds"; relay-summary.json "orbits"]. In the other networks each bond type is a single family.

Corollary 5 (A15 beyond Weaire and Phelan). Weaire and Phelan already used this family: as reported by Sloane and Vaishampayan [15], their weighted Voronoi partition of A15 puts the walls between unlike sites on the plane $(X - E)\cdot(D - E) = \mu|D - E|^2$ and has the 12-faced and 14-faced cells for $0 < \mu < 3/5$ [16]; Kashyap and Neuhoff [17] gave its cell volumes. That plane is a power bisector, and in our units $\lambda = (5/4)(2\mu - 1)$, so $\mu = 0, 1/2, 3/5$ are $\lambda = -5/4$ (the lower end), 0 (the polyhedral Weaire–Phelan cells, before Weaire and Phelan relaxed them into a curved foam) and 1/4 (the first flip); rescaled, Kashyap and Neuhoff's volume is the Si cell volume $(4\lambda + 5)^3/128$ on $(-5/4, 1/4)$ [S, prior-art.md, find 1]. What the census adds is the rest of the walk: on $(1/4, 3/4)$ Si–Si bonds exist alongside Cr–Cr; at $\lambda = 3/4$ the Cr–Cr bonds (both families) die exactly where a second Si–Si family is born, the Cr cells are tetrahedra with only unlike bonds and the Si cells have 20 triangles, with 12 corners of degree 5 (the face pattern of the icosahedron); at 5/4 the Si cells are truncated octahedra, the Voronoi cells of the body-centred cubic Si sites. Because the first Si–Si family has been alive since 1/4, $\lambda = 3/4$ is not a switch point.

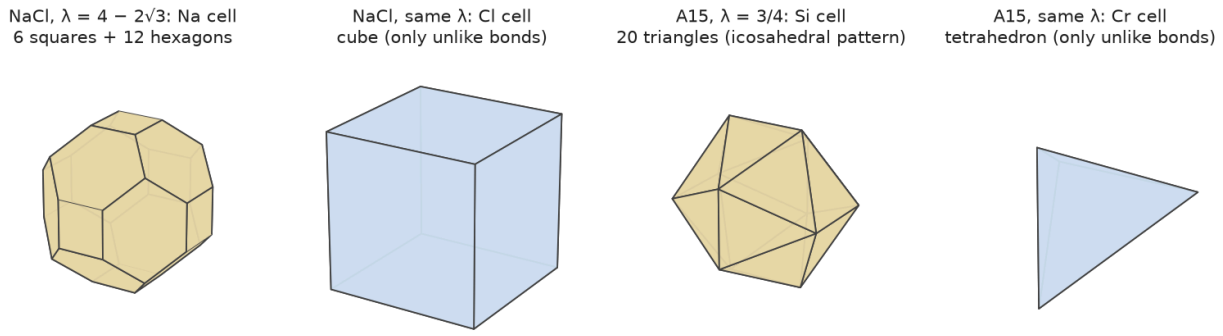


Figure 3. Two stops in space (float pictures of the exact cells). Left: rock salt at $\lambda = 4 - 2\sqrt{3}$, where every edge of the tiling has one length: the Na cell has 6 squares and 12 hexagons, the Cl cell is a cube. Right: A15 at the flip $\lambda = 3/4$: the Si cell has 20 triangles with 12 corners of degree 5, combinatorially an icosahedron, and the Cr cell is a tetrahedron with only unlike bonds.

Corollary 6 (perovskite). In the three-kind perovskite the two like bonds A–A and X–X are both present exactly on $0 < \lambda < 1 - \mu$, and their two lines meet only at the corner (0, 1) of the region, where the X cells vanish (Figure 4). So the hand-over is gradual throughout, and would only become a single switch at the edge of the region. At (0, −1) the B cells are points and the A and X cells are rhombic dodecahedra, the Voronoi cells of the face-centred cubic A + X sites.

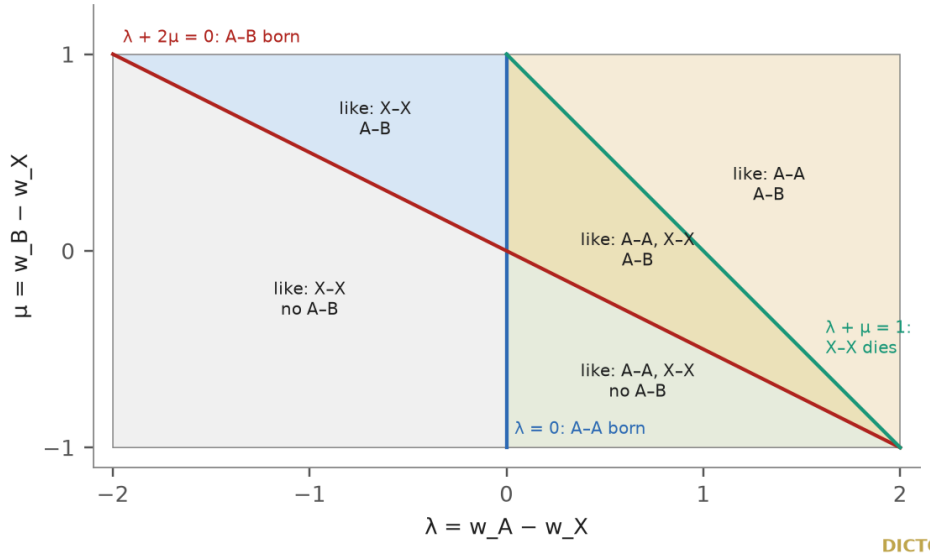


Figure 4. The perovskite chamber map (Main theorem (v)): the coexistence rectangle (open; its edges are where one kind has shrunk to points), its three walls and five chambers, each labelled with its like bonds and whether A–B is present. A–X and B–X are present everywhere; B–B never.

5. Observations and open problems

> *Some patterns hold in every network we computed, but we have not proved them in general. We state them as observations.*

Observation 1 (the diagonal swap). At each of the four switch points every like pair bonded on either side is a diagonal of a quadrilateral of the dual cell (a dual square, a dual rhombus, a face of a dual cube, a rhombus of a dual rhombic dodecahedron), and the switch trades one set of diagonals for the other in a single flip (Corollary 2; Figure 2; #52, Figure

3). This is read off the exact dual cells of these four cases. *Open problem:* is every switch point in every two-kind network a swap of like diagonals in dual quadrilaterals?

Observation 2 (CsCl and zinc blende). On each of the three intervals $(-3, -1)$, $(-1, 1)$ and $(1, 3)$, every bond-face area and every cell volume in zinc blende is the same polynomial in λ as in CsCl up to a constant factor: unlike faces 4 times larger, like faces $1/\sqrt{2}$ times, volumes twice; the flips (± 1) and the ends (± 3) are the same. For example, on $(-1, 1)$ the like bonds have area $(1 \pm \lambda)^2/(2\sqrt{2})$ in zinc blende against $(1 \pm \lambda)^2/2$ in CsCl, and the unlike bonds $\sqrt{3}(3 - \lambda^2)$ against $(\sqrt{3}/4)(3 - \lambda^2)$. The face counts differ (16 against 14 on $(-1, 1)$). In both networks the unlike nearest squared distance is 3, but the second kind does not sit in the same kind of hole: in CsCl it sits at the deep hole of the simple cubic lattice, in zinc blende at a tetrahedral hole of the face-centred cubic lattice, which is a shallow hole (the deep hole of fcc is the octahedral one) [18]. *Remark (ours, not an explanation):* in the units used here the zinc blende sites are exactly half of the CsCl sites (Zn: the points $2(a, b, c)$ of $2\mathbb{Z}^3$ with $a + b + c$ even, that is $2D_3$; S: those plus $(1, 1, 1)$). We have no explanation of the coincidence.

Observation 3 (which networks switch). A single switch occurs in square + centres, honeycomb + centres, rock salt and fluorite; a gradual hand-over in CsCl, zinc blende, the hcp holes, A15 and C15; kagome + centres has only one like bond. In two of the four switching networks the switch is at equal weights, where the sites form a lattice with non-simplicial Delaunay cells; in the other two it is not. We have no criterion that predicts the type from the sites.

Observation 4 (the ends of the range). In nine of the ten networks the coexistence range is $(-d^2, d^2)$, with d the nearest unlike distance ($d^2 = 3, 3/8, 2, 1, 1, 1, 3, 5/4, 11$ in the units of Table 1). The exception is the hcp holes: the upper end is $3/8 = d^2$, the lower end $-11/24$ rather than $-3/8$. The reason is exact: a tetrahedral hole of hcp has only three octahedral holes at squared distance $3/8$ (its fourth neighbour, across the shared face, is the other tetrahedral hole at $1/6$), so its cell does not shrink onto its own site but onto the centre $(0, 0, c/2)$ of the triangle shared by the two face-sharing tetrahedral holes. That point is at squared distance $1/24$ from the tet site and $1/3 + 1/6 = 1/2$ from the three nearest oct sites; equal power, $1/24 - \lambda = 1/2$, gives $\lambda = -11/24$. In general the end of the range is $\pm d^2$ exactly when the vanishing cell shrinks onto its own site; in fluorite each tet site has four oct neighbours around it and does.

6. Conclusion

> *DICTO's proposal holds: both branches of the DICTONIC loops, the gradual hand-over and the single switch, recur in classical networks, in slightly different guises.*

The single switch of the Delian loop reappears in the square lattice with its centres, in the honeycomb with its hexagon centres (where the switch is the trihexagonal tiling) and in rock salt (where it is the cubic honeycomb of equal weights); the gradual hand-over of the golden loop reappears in zinc blende, A15, C15 and the holes of hexagonal close packing. The bond between unlike pieces never breaks in any of them, and Proposition 1 shows it cannot. The golden loop is exactly the gradual stretch of the caesium chloride family, and the Delian loop is the whole fluorite family. One guise is new to the DICTONIC picture: in hexagonal stacking the octet's flip survives at the same weight but stops being a switch point, because the

gyrated layers keep like bonds across them, and a second flip follows at $\lambda = 5/24$. Open: Observations 1–4, the full exact walk of other three-kind networks, and whether the switch-point networks share a common reason.

7. What is known, and what was not found

> *Power diagrams, their flips, the named tilings and the A15 family up to its first flip are known. We did not find the full walks of these networks.*

Known. Power (Laguerre) diagrams and their duality with regular subdivisions [6, 7]; regular subdivisions change only at the walls of the secondary fan, where non-simplicial cells are cut in two ways [8, 9], including for periodic sets [19] and lattices (Voronoi's L-types [10, 11]); the circumsphere rule and the criteria for a tiling to be a power diagram [12, 13, 14]; the uniform tilings and honeycombs at the stops [20, 21, 22]; the classical Voronoi cells at equal weights and at the ends (squares, hexagons, triangles, rhombille rhombi; cubes, rhombic dodecahedra, truncated octahedra, the triakis truncated tetrahedra of diamond, hexagonal prisms) [18, 20]; the Weaire–Phelan cells as the A15 Voronoi cells and the A15 weighted family for $0 < \mu < 3/5$, our $\lambda \leq 1/4$, with its cell volumes [15, 16, 17]; the C15 cells [23, 24]; radical tessellations of binary crystals, including one point of the rock salt interval $(0, 1)$: a crystal of large spheres on the face-centred cubic lattice with small spheres in the octahedral pores, whose large-sphere radical polyhedra have 6 quadrangles and 12 hexagons and whose small-sphere polyhedra are cubes [25]; Dirichlet-domain types of cubic lattice complexes changing at critical positional parameters (unweighted; the closest crystallographic analogue) [26]; the two DICTONIC loops as power diagrams [3]; the gyrated octet in DICTO's earlier work [4, 5].

Not found, in the sources listed in Appendix A: the full weight walks of square + centres, honeycomb + centres, kagome + centres, rock salt, zinc blende, the hcp holes, A15 beyond $\lambda = 1/4$, C15 and the perovskite, with their exact flips, bond lives and switch points; the split of these networks into gradual and single-switch hand-overs; the switch point of honeycomb + centres at the trihexagonal tiling; the second flip of the hcp holes at 5/24; the CsCl–zinc blende coincidence; the rock salt stop $\lambda = 4 - 2\sqrt{3}$. Several are elementary and may well be known in the radical-tessellation or crystallographic literature; Koch 1973 [26] and the radical-tessellation papers were not read in full. "Not found" is limited to those sources.

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Appendix A. Sources checked for prior work

The prior-art search of 25 October 2026 (file prior-art.md in the study folder [S]), on top of #52's [3]. DICTO's reference library (kaleidohedra-references): library searches on power diagrams, Laguerre and radical tessellations of rock salt and zinc blende, weighted Voronoi diagrams of Archimedean tilings, and Voronoi cells of lattice complexes (Koch, Fischer); its index searched for Laguerre, power diagram, Aurenhammer, weighted Voronoi, Frank–Kasper, Weaire, Laves, A15, zinc blende, trapezo, radical, chamfered, trihexagonal, Tilings and Patterns, Andreini, Koch, Fischer, Sloane, O'Keeffe, Hyde, Blatov, Dolbilin, Engel, Dirichlet, lattice complex, Archimedean, Grünbaum. Koch 1973 [26], Blatov 2004, O'Keeffe 1979, Frank and Kasper 1958 [23] and Conway, Burgiel and Goodman-Strauss [22] are in the library as links only. Web: about 12 queries (radical, Laguerre and power tessellations with rock salt, zinc blende, perovskite and binary hard-sphere crystals; Archimedean tilings as power diagrams; Fischer, Koch and Hellner; Frank–Kasper and Laves Voronoi cells; Weaire–Phelan weighted Voronoi; Kusner–Sullivan; O'Keeffe and Hyde; weighted Delaunay flips in crystals; the trihexagonal tiling as a power diagram). Read in full: Sloane and Vaishampayan [15] (arXiv). Read in part: Wang et al. [25] (publisher page and its table). Every DOI was checked on Crossref (title, venue, volume, pages, year). **Not read:** the full

texts of Koch 1973 [26], Weaire and Phelan [16], Kashyap and Neuhoff [17] (their formula is quoted from [15]), Ash and Bolker [13, 14], Blatov 2004, Engel's Geometric Crystallography, O'Keeffe and Hyde's Crystal Structures; Google Scholar itself. Reddy et al. [27] describe A15 as two 12-faced and six 14-faced Voronoi cells and use weighted partitions of Frank–Kasper phases; no C15 weight walk was found there.

Appendix B. Checks

Part	What is checked	PASS	FAIL
walks (census.py)	per interval: volume sum identity, face to face, Euler, distance bound; per stop: limit = cell from scratch; classical cells. square 13, honeycomb 18, kagome 11, CsCl 25, fluorite 58, rock salt 42, zinc blende 61, hcp holes 48, A15 60, C15 207	543	0
bond lives (relay.py)	each bond type on one interval; single edge length at the planar switch points; 9 regressions against #52's study	41	0
perovskite (perovskite.py)	chambers: distance bound, Euler, volume sum, face to face; bounded region; every wall covered; rhombic dodecahedra at (0, −1)	47	0
cross-check (crosscheck.py)	float cells at 3 random parameters per interval (10 networks); exact dual cells at the two octet-type stops	12	0
perovskite walls (walls.py, added after review)	exact cells at the midpoints of the five interior wall segments: all kinds alive, a like bond present	5	0
total		**648**	**0**

Exact arithmetic: Python fractions and sympy (polynomials in λ, or in λ and μ); a number is zero only if it is exactly zero; floats only choose candidate corners and serve as the independent cross-check. Run on one machine, one job at a time, about 3.5 minutes in all [S, README "Run times"].

Appendix C. The walks in full

Generated from the study's data/raw/tables.md [S] when this paper is built (not retyped). For every interval (in brackets) and every stop: the bond types present, and for each kind its cell (face vector: 8×3+6×4 means 8 triangles and 6 squares; in the plane, n-gon) and its bonds by neighbour kind; "point" means the cell has shrunk to a point.

square lattice + centres · $\lambda = w_{corner} - w_{centre}$

λ	bond types	centre cell (faces)	centre bonds	corner cell (faces)	corner bonds
−2	centre–centre	4-gon	4 centre	point	
(−2, 0)	centre–centre, centre–corner	8-gon	4 centre, 4 corner	4-gon	4 centre
0	centre–corner	4-gon	4 corner	4-gon	4 centre
(0, 2)	centre–corner, corner–corner	4-gon	4 corner	8-gon	4 centre, 4 corner
2	corner–corner	point		4-gon	4 corner

honeycomb + hexagon centres · $\lambda = w_{honeycomb} - w_{centre}$

λ	bond types	centre cell (faces)	centre bonds	honeycomb cell (faces)	honeycomb bonds
−1	centre–centre	6-gon	6 centre	point	
(−1, −1/2)	centre–centre, centre–honeycomb	12-gon	6 centre, 6 honeycomb	3-gon	3 centre
−1/2	centre–honeycomb	6-gon	6 honeycomb	3-gon	3 centre
(−1/2, 1)	centre–honeycomb, honeycomb–honeycomb	6-gon	6 honeycomb	6-gon	3 centre, 3 honeycomb
1	honeycomb–honeycomb	point		3-gon	3 honeycomb

kagome + hexagon centres · $\lambda = w_{kagome} - w_{centre}$

λ	bond types	centre cell (faces)	centre bonds	kagome cell (faces)	kagome bonds
−1	centre–centre	6-gon	6 centre	point	
(−1, 1)	centre–kagome, kagome–kagome	6-gon	6 kagome	6-gon	2 centre, 4 kagome
1	kagome–kagome	point		4-gon	4 kagome

CsCl (bcc two-kind; golden-loop sites) · $\lambda = w_{big} - w_{small}$

λ	bond types	big cell (faces)	big bonds	small cell (faces)	small bonds
−3	small–small	point		6×4	6 small
(−3, −1)	big–small, small–small	8×3	8 small	8×3+6×8	8 big, 6 small
−1	big–small, small–small	8×3	8 small	8×3+6×4	8 big, 6 small
(−1, 1)	big–big, big–small, small–small	6×4+8×6	6 big, 8 small	6×4+8×6	8 big, 6 small
1	big–big, big–small	8×3+6×4	6 big, 8 small	8×3	8 big
(1, 3)	big–big, big–small	8×3+6×8	6 big, 8 small	8×3	8 big
3	big–big	6×4	6 big	point	

fluorite (fcc + all tetrahedral holes; Delian-loop sites) · $\lambda = w_{tet} - w_{oct}$

λ	bond types	oct cell (faces)	oct bonds	tet cell (faces)	tet bonds
−3/8	oct–oct	12×4	12 oct	point	
(−3/8, −1/8)	oct–oct, oct–tet	8×3+12×6	12 oct, 8 tet	4×3	4 oct
−1/8	oct–tet	8×3	8 tet	4×3	4 oct
(−1/8, 3/8)	oct–tet, tet–tet	8×3	8 tet	4×3+6×6	4 oct, 6 tet
3/8	tet–tet	point		6×4	6 tet

rock salt NaCl · $\lambda = w_{Na} - w_{Cl}$

λ	bond types	Cl cell (faces)	Cl bonds	Na cell (faces)	Na bonds
−1	Cl–Cl	12×4	12 Cl	point	
(−1, 0)	Cl–Cl, Cl–Na	6×4+12×6	12 Cl, 6 Na	6×4	6 Cl
0	Cl–Na	6×4	6 Na	6×4	6 Cl
(0, 1)	Cl–Na, Na–Na	6×4	6 Na	6×4+12×6	6 Cl, 12 Na
1	Na–Na	point		12×4	12 Na

zinc blende ZnS · $\lambda = w_{Zn} - w_{S}$

λ	bond types	S cell (faces)	S bonds	Zn cell (faces)	Zn bonds
−3	S–S	12×4	12 S	point	
(−3, −1)	S–S, S–Zn	4×3+12×5	12 S, 4 Zn	4×3	4 S
−1	S–S, S–Zn	16×3	12 S, 4 Zn	4×3	4 S
(−1, 1)	S–S, S–Zn, Zn–Zn	12×3+4×6	12 S, 4 Zn	12×3+4×6	4 S, 12 Zn
1	S–Zn, Zn–Zn	4×3	4 Zn	16×3	4 S, 12 Zn
(1, 3)	S–Zn, Zn–Zn	4×3	4 Zn	4×3+12×5	4 S, 12 Zn
3	Zn–Zn	point		12×4	12 Zn

hcp holes (octahedral + tetrahedral holes of ideal hcp) · $\lambda = w_{tet} - w_{oct}$

λ	bond types	oct cell (faces)	oct bonds	tet cell (faces)	tet bonds
−11/24	oct–oct	6×4+2×6	8 oct	point	
(−11/24, −1/8)	oct–oct, oct–tet, tet–tet	6×3+6×6+2×9	8 oct, 6 tet	4×3	3 oct, 1 tet
−1/8	oct–oct, oct–tet, tet–tet	8×3	2 oct, 6 tet	4×3	3 oct, 1 tet
(−1/8, 5/24)	oct–oct, oct–tet, tet–tet	2×3+6×5	2 oct, 6 tet	6×3+3×5+3×6+1×9	3 oct, 10 tet
5/24	oct–tet, tet–tet	6×4	6 tet	6×3+3×4+4×6	3 oct, 10 tet
(5/24, 3/8)	oct–tet, tet–tet	6×4	6 tet	9×4+4×6	3 oct, 10 tet
3/8	tet–tet	point		6×3+3×4+1×6	10 tet

A15 Cr_3Si · $\lambda = w_{Si} - w_{Cr}$

λ	bond types	Cr cell (faces)	Cr bonds	Si cell (faces)	Si bonds
−5/4	Cr–Cr	8×3+2×4	10 Cr	point	
(−5/4, 1/4)	Cr–Cr, Cr–Si	12×5+2×6	10 Cr, 4 Si	12×5	12 Cr
1/4	Cr–Cr, Cr–Si	8×4+4×5+2×6	10 Cr, 4 Si	12×5	12 Cr
(1/4, 3/4)	Cr–Cr, Cr–Si, Si–Si	8×4+2×6+4×7	10 Cr, 4 Si	8×3+12×7	12 Cr, 8 Si
3/4	Cr–Si, Si–Si	4×3	4 Si	20×3	12 Cr, 8 Si
(3/4, 5/4)	Cr–Si, Si–Si	4×3	4 Si	12×3+6×6+8×9	12 Cr, 14 Si
5/4	Si–Si	point		6×4+8×6	14 Si

C15 Laves $MgCu_2$ · $\lambda = w_{Mg} - w_{Cu}$

λ	bond types	Cu cell (faces)	Cu bonds	Mg cell (faces)	Mg bonds
−11	Cu–Cu	6×4	6 Cu	point	
(−11, −5)	Cu–Cu, Cu–Mg	6×3+6×7	6 Cu, 6 Mg	12×3	12 Cu
−5	Cu–Cu, Cu–Mg	6×3+6×5	6 Cu, 6 Mg	12×3	12 Cu
(−5, 7)	Cu–Cu, Cu–Mg, Mg–Mg	12×5	6 Cu, 6 Mg	12×5+4×6	12 Cu, 4 Mg
7	Cu–Cu, Cu–Mg, Mg–Mg	12×4	6 Cu, 6 Mg	12×4+4×6	12 Cu, 4 Mg
(7, 9)	Cu–Cu, Cu–Mg, Mg–Mg	6×4+6×6	6 Cu, 6 Mg	12×3+12×6+4×12	12 Cu, 16 Mg
9	Cu–Mg, Mg–Mg	6×4	6 Mg	12×3+12×4+4×12	12 Cu, 16 Mg
(9, 11)	Cu–Mg, Mg–Mg	6×4	6 Mg	12×4+12×5+4×12	12 Cu, 16 Mg
11	Mg–Mg	point		12×3+4×6	16 Mg